REMORA: Real-time Expressive Motion Routing for Audio

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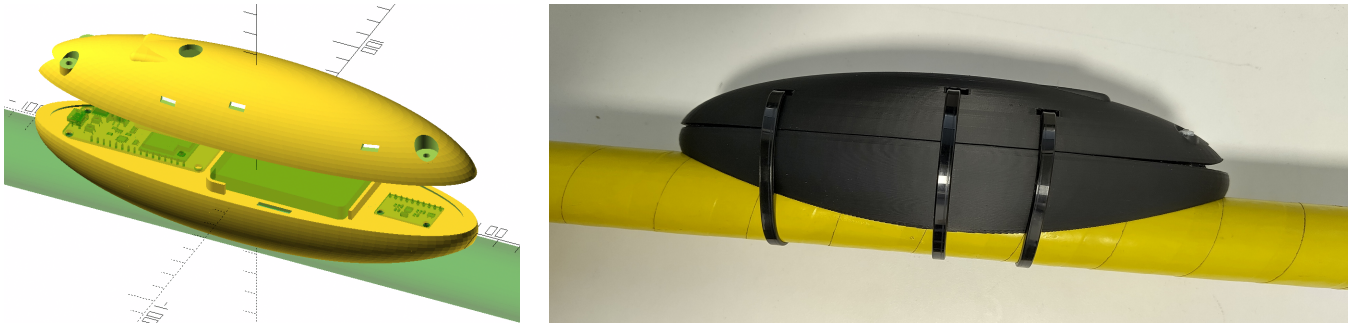


Figure 1: The REMORA sensor unit: parametric OpenSCAD enclosure (left) and the printed enclosure zip-tied near one end of the bô (right).

Abstract

REMORA turns a martial-arts staff (bô) into a musical instrument. A wireless inertial sensor on the staff streams its orientation and motion to an audio plugin, where spins, changes of facing, and thrusts become sound in real time. The bô is a full-length, two-handed, swung object with its own rotational inertia and an established movement vocabulary, and we derive features specific to it: the plane and direction of a spin, a coarse facing that needs no magnetometer, a lap counter, and an axial thrust detector. The gesture-to-sound mapping is a small dataflow patch that the performer can see and rewire on a canvas inside the plugin while it plays. We describe the hardware, a user-guided calibration that makes the instrument independent of how the sensor is rotated on the staff, the motion features, and the mapping layer, and we discuss how common bô techniques relate to the presets. A pilot session with ten first-time players found the instrument comfortable and engaging, and identified thrust detection as its main defect, for which we give a probable cause.

CCS Concepts

• **Applied computing** → **Sound and music computing**;
• **Software and its engineering** → *Visual languages.*

Keywords

digital musical instrument, martial arts, bô staff, inertial sensing, gesture mapping, visual patching, OSC

1 Introduction

1.1 Motivation

REMORA comes from the first author's practice in both martial arts and music. The bô is a staff of about 1.5 m, held in two hands and swung, spun, and thrust through a body of techniques that are taught as forms (*kata*). Inertial-sensor Digital Musical Instruments (DMIs) presented at New Interfaces for Musical Expression (NIME) are mostly small handheld or body-worn objects. A bô instead brings a full-body object, rotational inertia, reach, and an existing gesture vocabulary that a trained performer already controls precisely.

Finding a gesture vocabulary for such an object took many attempts. At first each gesture-to-sound mapping was its own code module, and every change meant a rebuild. After eleven of them, we moved the mapping into a small dataflow patch that is edited live inside the audio plugin that makes the sound. Visual patching is well established (§2); in REMORA it is the tool for writing and revising the staff's vocabulary.

1.2 Contributions

This paper describes a working instrument and a first pilot study. Its contributions are:

(1) A bô-staff instrument whose motion features are specific to a long swung object: spin plane and direction, a coarse facing derived without a magnetometer, a continuous lap count, and an axial thrust detector (§4.3).
(2) A two-stage, user-guided calibration that removes the effect of how the sensor is rotated on the staff (§4.2).
(3) A mapping layer inside the plugin, built as a typed directed acyclic graph (DAG) that the performer can rewire while audio runs (§4.4).
(4) Fifteen presets built from one catalog of 69 node types, from a two-control starter mapping to spin- and facing-driven mappings for trained players (§4.5).

Table 3 sets out the evidence for each contribution and what remains untested.

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1.3 Scope

The intended performer is a soloist with some staff or martial-arts background, playing a set built from presets. The catalog is graded: *Simple* needs only tilt and roll, while the facing- and spin-driven *Martial Momentum* and *Azimut Kinetic* reward bô technique. Because the mapping is a patch, REMORA may also interest patch-based electronic musicians looking for a full-body controller. Ensemble use is out of scope.

2 Related Work

2.1 Rod, Baton, and Swung Controllers

The *T-Stick* [10] senses touch and pressure along its shaft together with accelerometer-derived shake and tap gestures. Conducting interfaces include Mathews' *Radio Baton* [13], Marrin and Paradiso's *Digital Baton* [11], and the *Conductor's Jacket* [12]. Nam's *mPoi* [14] established swung objects as a DMI form, and *x-OSC* [8] popularised the Wi-Fi/OSC inertial streaming pattern that REMORA's sensor unit follows.

2.2 Mapping Tools and Visual Patching

Visual dataflow patching descends from Puckette's Patcher, now *Max* [16, 18], and continues in *Pure Data* [17] and *VCV Rack* [1]. For DMIs, libmapper [9] and its graphical front end Webmapper [23] connect controller signals to synthesis parameters across a network, and *ossia score* [3] combines such mappings with a timeline. Any of these could host REMORA's mappings. We put the graph inside the plugin for practical reasons: mapping and synthesis share one process and one preset file, and the patch shows each signal's live value next to the synth it drives (§4.6). The closest prior system is the T-Stick with libmapper. REMORA differs in its controller and features: a full-length swung bô has a spin plane, a spin direction, and a facing, for which a handheld rod's touch and shake gestures have no counterpart.

2.3 Mapping Strategies

Hunt and Wanderley [7] classify mappings as one-to-one, one-to-many, or many-to-one, and argue that richer mappings make an interface feel like an instrument. Van Nort et al. [21] extend mapping to continuous navigation between parameter spaces; REMORA's graph stays with explicit parameter-level functions. *Wekinator* [5] and Françoise et al. [6] learn mappings from example gestures instead of having the performer wire them.

Expressive-movement systems often describe motion with Laban's Effort factors [2, 19], and Fdili Alaoui et al. [4] show that capturing them well takes expert annotation and multimodal data. REMORA uses simpler kinematic signals tied to the staff (§4.3).

3 Design

Five requirements shaped the design:

- **A real bô.** The instrument keeps the reach, weight, and two-handed grip of a full-length staff.
- **Wireless during performance.** A cable is acceptable only for charging.
- **Live-editable mapping.** Loading or rewiring a mapping must not interrupt audio.
- **Mount-agnostic calibration.** Behaviour should not depend on exactly how the sensor is strapped to the staff.
- **Low latency.** The sound should follow the motion closely enough to feel driven by it (§6.2).

The firmware samples the Inertial Measurement Unit (IMU) and sends Open Sound Control (OSC) packets over Wi-Fi. The plugin does everything downstream: OSC reception, calibration, motion analysis, graph evaluation, and synthesis.

4 Implementation

4.1 Hardware

The sensor unit is a SparkFun ESP32-S2 Thing Plus reading an MPU-9250 IMU over I2C at 400 kHz, powered by a 2000 mAh LiPo battery (Figure 2). It sits in a parametric OpenSCAD enclosure sized to the staff diameter (Figure 1) and is attached near one end, out of the performer's grip, with zip ties, so it can be moved or removed without damaging the staff. Because calibration absorbs the sensor's rotation, the enclosure can be strapped on at any angle. Its position along the staff does matter (§4.3) and should be kept constant between sessions.

Orientation is estimated on the microcontroller by the `hideakitai/MPU9250` library (v0.4.8) with its Mahony filter, which fuses accelerometer and gyroscope only; the magnetometer is not used (§7.1). The library's default ranges (±16 g, ±2000°/s) are kept; fast spins approach the gyroscope limit. Each OSC packet carries the quaternion, raw acceleration, angular rate, and magnetic field, and is sent over UDP at most once every 10 ms (nominally 100 Hz). The host computer runs a Wi-Fi hotspot and the staff joins it directly, so no separate access point is needed.



Figure 2: Inside the enclosure: ESP32-S2, MPU-9250 IMU, and LiPo battery.

4.2 Calibration

Orientation is tracked as a unit quaternion; rotating v by q gives $v' = q\,v\,q^{-1}$. The performer is prompted through three poses, in order: A, staff horizontal pointing forward; B, staff vertical pointing up; C, staff horizontal pointing right (quaternions q_A, q_B, q_C).

Stage 1: which sensor axis runs along the staff. Each candidate axis $L \in \{\pm X, \pm Y, \pm Z\}$ is scored by

$$v_P = \widehat{\text{rotate}(L, q_P)}\,, \qquad e(L) = |v_A \cdot v_B| + |v_B \cdot v_C| + |v_C \cdot v_A|\,.$$

Only the staff axis is carried through three mutually orthogonal directions by the three poses, so it has the lowest $e(L)$. L and $-L$ score the same but give opposite signs of the triple product $(v_A \times v_B) \cdot v_C$. Which sign marks the tip depends on the IMU library's frame convention; with the library we use, it is the negative one:

$$L^* = \operatorname*{arg\,min}_{L\,:\,(v_A \times v_B)\cdot v_C < 0} e(L)\,,$$

which gives the alignment quaternion q_{align}. If the poses are too close together or nearly coplanar, calibration is rejected and the performer is asked to retry.

Stage 2: heading and twist. From the measured staff axis in poses A and B (b_0, b_1), an orthonormal triad $U = [\, b_0,\ \widehat{b_0 \times b_1},\ b_0 \times \widehat{b_0 \times b_1} \,]$ is built, and likewise W from the instrument-frame forward $(1, 0, 0)$ and up $(0, 0, 1)$. The aligning rotation is the orthogonal-Procrustes solution $R = WU^{\top}$, converted to a quaternion q_{corr}. Every reading is then mapped to instrument space by $q' = q_{\text{corr}}\,(q_{\text{raw}}\,q_{\text{align}})$, whatever the sensor's rotation on the staff. Its position along the staff is not corrected (§7.1).

4.3 Motion Features

A single motion analyzer computes the features in Table 1, and every mapping graph reads from it. It currently runs once per audio block on the latest sensor packet, with Δt from the host clock (§7.1). All smoothing rates and thresholds below were tuned by ear on the instrument, at one host buffer size.

Table 1: Motion features computed by the analyzer.

Feature	Purpose
Tip vector, rotation axis	tip velocity and spin axis
Dynamic acceleration	motion with gravity removed
Energy, onset, axis stability, jerk	how hard and how steadily the staff moves
Spin plane and direction	vertical/horizontal spin, CW/CCW
Facing	which horizontal axis the staff aligns with
Continuous spin count	full turns in the current spin
Axial thrust	stab or strike along the staff

Tip, axis, and gravity. The tip direction $p_n = \text{rotate}((1, 0, 0), q_n)$ is kept over three samples to estimate tip velocity and the current rotation axis. Gravity is estimated by smoothing the inverse-rotated world $+Z$ axis with a one-pole filter (α=0.10), and subtracted from the raw reading to give the dynamic acceleration a_{dyn}.

Energy features. Four smoothed signals describe how hard the staff is moving. Each applies the one-pole smoother $y_n = y_{n-1} + \alpha\,(x_n - y_{n-1})$ to a derivative of a_{dyn} or of gyroscope magnitude. *Energy* is the enveloped magnitude of a leaky integral of a_{dyn}. *Onset* is the positive rate of change of gyroscope magnitude. *Axis stability* is the smoothed absolute dot product between consecutive rotation axes, high when the staff keeps spinning around the same axis. *Jerk* is the smoothed jerk of $|a_{\text{dyn}}|$, split into smooth and abrupt parts. (In the source code these four are named after Laban's Effort factors: Weight, Time, Space, and Flow.) With the unit near the tip, the sensor sits about $r \approx 0.7$ m from the centre of rotation, and a 2 rev/s spin produces a centripetal acceleration $\omega^2 r \approx 11$ g. This term dominates a_{dyn}, so Energy and Jerk mainly follow spin speed, which is how the presets use them, and their scale changes if the unit is moved along the staff.

Spin plane and direction. Let (a_x, a_y, a_z) be the rotation axis, smoothed with α=0.35. The spin is classified as *vertical* (the staff turns in a vertical plane, like a baton twirled beside the body) when

$$\sqrt{a_x^2 + a_y^2} > |a_z|\,,$$

and as *horizontal* (like a propeller overhead) otherwise. For a horizontal spin, the direction is the sign of a_z. For a vertical spin, it is the sign of the larger of a_x, a_y, or, in the *Martial* presets, the sign of the axis' dot product with a reference axis latched at the first vertical spin after the preset loads.

Facing. A compass heading would need the magnetometer, which is unreliable next to the staff's electronics and stage equipment. REMORA instead asks which horizontal axis of the calibrated frame the staff is closer to: x (the calibration "forward", labelled *north*) or y (*east*). Let (d_x, d_y) be the horizontal components of the rotation axis during vertical spins, or of the tip direction otherwise. With hysteresis factor $k = 1.15$, the state switches

$$\text{north} \to \text{east if } |d_y| > k\,|d_x|\,, \qquad \text{east} \to \text{north if } |d_x| > k\,|d_y|\,.$$

Measured from the x axis, the switch points are therefore $\arctan(1.15) \approx 49°$ and $\arctan(1/1.15) \approx 41°$, which prevents flicker at the boundary. Because orientation is 6-axis, heading is relative to the calibration pose and drifts with gyroscope yaw (§7.1).

Continuous spin count. $\|\omega\|\,\Delta t$ is accumulated, and a lap is counted every 360°; the count resets when the spin plane or direction changes.

Axial thrust. A thrust is flagged when the jerk of the dynamic acceleration projected on the staff axis crosses zero while its magnitude exceeds 1.5 g, gated to slow rotation ($\|\omega\| < 90°$/s) and followed by a cooldown.

4.4 Mapping Architecture

The node catalog has four categories: Source (sensor values and the features above), Math, Sink (one node per synthesis engine exposing all its parameters as typed inputs, plus output gain), and Display (live monitors). A graph is a set of nodes plus typed wires, saved as an XML preset. The graph is the only path from sensor to synthesiser.

When a wire is added, the graph recomputes a topological order with Kahn's algorithm and refuses the wire if it would close a cycle. Each audio block, evaluation walks the cached order once and every node computes $y_i = f_{\text{type}(n_i)}(x_{i,1}, \ldots, x_{i,m}; \theta_i)$ from its inputs x and parameters θ. The patch is thus one composed function of the sensor frame, with no sorting on the audio thread. If the user interface is editing the graph when a block arrives, evaluation does not wait; it reuses the previous block's output.

Math nodes cover arithmetic, shaping (linear and logarithmic range mapping, clamp, threshold, crossfade, quantise, sine, atan2, semitones to Hz), lookup tables (1-D, and an 8-entry table indexed by three booleans), and stateful dynamics (one-pole smoother, leaky integrator, retrigger envelope, hysteresis, derivative, latch, random sample-and-hold).

4.5 Presets

The eleven C++ mappings from development were rebuilt as graph presets: *Simple*, *Lead+Drone*, *Spin Filter*, *Bowed Chord*, four *Azimut* variants, *Martial Effort*, *Martial Momentum*, *Spin Voices*, and *Speed Gate*. Four more (*Wind Noise*, *Azimut Kinetic*, *Spin Voices (Scale)*, *Vocal Tract*) were made directly as graphs. Nine presets are offered in the plugin's selector (Table 2); the others are close variants kept as reference patches.

Table 2: The nine presets in the plugin's selector.

Preset	Character
Simple	Pitch from tilt, volume from roll; one sine voice.
Wind Noise	No pitch; motion energy swells filtered noise into gusts.
Bowed Chord	Root from tilt; yaw and roll pick the chord; spin speed acts like bow pressure.
Lead+Drone	Major-scale lead over a 4-voice drone.
Martial Momentum	Root from spin plane and direction; glide speed from spin momentum.
Azimut Kinetic	Root from spin and facing; rotation speed drives the whole timbre.
Speed Gate	Slow: pentatonic melody. Fast: crossfades to the Azimut mapping.
Spin Voices (Scale)	Roll selects 1 of 4 scale-quantised voices.
Vocal Tract	Physically modelled voice; jabs add consonant-like bursts.

The catalog has 69 node types (33 source, 29 math, 4 sink, 3 display). Presets range from 7 nodes (*Simple*) to 178 (*Martial Effort*), with a median of 86. Together they use 50 of the 69 types; most unused types are raw per-axis sensor values. No preset needed a node type of its own. Since the catalog grew alongside the hand-coded mappings, this shows that it can express them; mappings designed by someone who did not build the catalog are still to be tried. Per-preset counts are generated from the code by a tool in the repository.

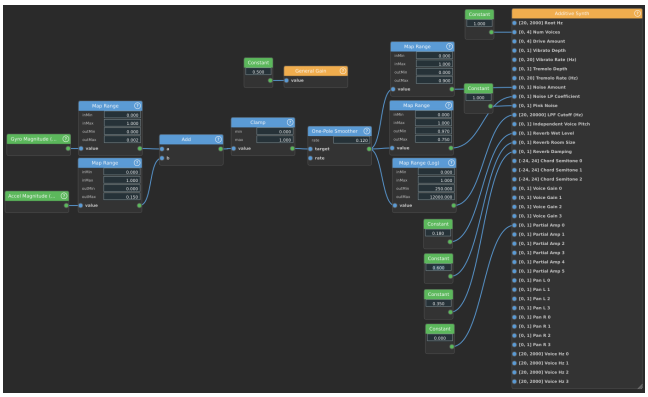


Figure 3: The complete *Wind Noise* preset in the plugin: gyroscope and acceleration magnitudes are scaled, summed, clamped, and smoothed into one motion-energy signal that drives the noise level, noise filter, and lowpass cutoff of the additive engine (right).

Azimut root table. The Azimut presets and the fast branch of *Speed Gate* choose the root note from three booleans (vertical spin, counter-clockwise, facing east) through the 8-entry lookup node. The target pitch is reached by a one-pole glide whose rate grows with rotation speed and jumps at motion onset, so faster motion moves to the new root more quickly (Figure 4).

Kinetic timbre. *Azimut Kinetic* keeps the Azimut root and maps one smoothed rotation-energy value $E \in [0, 1]$ (gyroscope magnitude normalised over 30–750°/s) to many timbre parameters at once [7]: lowpass cutoff, partial amplitudes, drive, noise, vibrato, tremolo, and reverb. A still staff sounds close to a pure sine; fast spins sound buzzy, noisy, and reverberant. Each completed lap also triggers a random sample-and-hold, whose values drift slowly into the same parameters, so two similar spins give two nearby but different timbres. The whole preset is built from existing math nodes.

4.6 Live Patching

The node-graph editor (Figure 4) is a pannable, zoomable canvas. Nodes are coloured by category, wires are drawn by dragging between typed pins, and new nodes come from a searchable, categorised picker. Hovering any pin shows its current value, and Display nodes (number, meter, scope) can be placed on a wire to monitor it. A changed patch can be kept or reverted to its saved preset in one click, so a mapping can be tried during rehearsal and then kept or dropped.

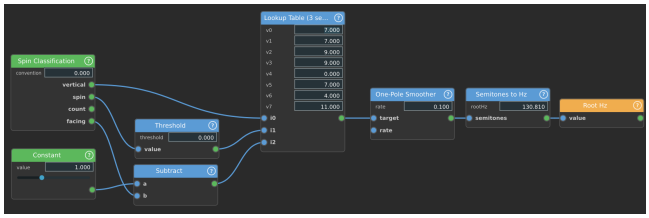


Figure 4: The node-graph editor showing the Azimut root-note chain: spin plane, direction, and facing select an entry of the 3-selector lookup table, which is smoothed and converted to the root frequency (green Source, blue Math, amber Sink).

4.7 Synthesis

Synthesis engines share one interface and are skipped entirely when the active preset does not wire them. Every preset except *Vocal Tract* drives the additive engine: four voices of six harmonic partials, with soft-clip waveshaping, vibrato, tremolo, chorus, filtered noise, a state-variable lowpass, and reverb. *Vocal Tract* drives a C++ port (via `cutelabnyc/pink-trombone-cpp`, MIT-licensed) of Neil Thapen's *Pink Trombone* [20]; only the control layer around it is ours. If no packet arrives for 150 ms, the master gain fades out over 120 ms, and fades back in over 10 ms when packets return.

4.8 Development Iterations

The graph came last. The first iteration streamed OSC to a minimal receiver with one hard-coded tilt-to-pitch mapping. The second wrote the mappings as separate C++ classes on top of the shared analyzer, each adding vocabulary (chords, energy features, facing, thrust). Read side by side, they were the same arithmetic, shaping, and stateful primitives wired in different patterns, so the third iteration replaced the classes with the graph engine. Since then, a second synthesis engine (§4.7) and four new presets have been added without changes to the engine.

5 Playing the Bô: From Technique to Sound

The staff-specific features were designed around families of bô technique, and this is where the presets differ most from a generic motion controller.

Spins. Two-handed spins beside the body and overhead spins are the staff's most characteristic motions. Spin plane and direction select among four roots in *Martial Momentum*, so switching from a vertical clockwise spin to a horizontal one is heard as a change of harmony, and the length of a spin, through the lap count, as a slow timbral sweep in the Azimut presets.

Changes of direction. A kata follows a floor pattern (*embusen*) with turns between facings. Facing adds a third boolean to the Azimut root table, so the same spin played after a quarter turn lands on a different root, and a form traced across the floor becomes a sequence of harmonic regions.

Stances and transitions. Held guard positions (*kamae*) are near-static and read as tilt and roll. In *Simple* and *Bowed Chord*, stance chooses pitch or chord; in *Azimut Kinetic*, stillness gives a near-pure tone and energy brightens it, so the contrast between a held stance and the motion out of it shapes the timbre directly.

Thrusts and strikes. Thrusts (*tsuki*) and strikes are short, decisive, and mostly along the staff axis. They are what the axial detector targets, and what pilot participants noticed was missing in the sound (§6.3).

6 Evaluation

The evidence consists of properties of the implemented system and an exploratory pilot. Table 3 summarises it per contribution.

6.1 Evidence from the Implemented System

Rewiring is exercised every time a preset loads or a wire changes. The rebuilt presets were compared with the original classes by ear; replaying one recorded IMU log through both is still to be done. Rotation independence of the calibration follows from its derivation, and a remount test at several angles on hardware is planned.

6.2 Latency

We estimate the end-to-end latency, from a movement of the staff to the corresponding sound, at around 50 ms. The chain is: IMU read and orientation update on the microcontroller, a send interval of up to 10 ms, Wi-Fi delivery to the host's hotspot, analysis and graph evaluation at the next audio block, and the host's output buffer. The figure has yet to be confirmed by measurement (§6.4). It is well above the 10 ms often cited for intimate control [24]. Most presets map continuous motion (spins, sweeps, the move out of a stance) to slowly gliding parameters, which tolerate such a delay better than percussive events do. Thrusts and strikes are the exception: their sound should land with the impact, so reducing latency, for instance with a shorter send interval or a dedicated radio link (§7.2), belongs with the thrust work in §7.1.

6.3 Pilot Session

Ten adults with no prior exposure to REMORA took part in a short, informal pilot in August 2026, using the current firmware (see Ethical Statement). Each was calibrated, given a brief demonstration, and then played freely across a subset of the nine selector presets before completing an anonymous questionnaire. Nine of ten rated their martial-arts or staff experience 1 or 2 out of 5; general musical experience averaged 3.4/5, and prior experience with gestural instruments 2.6/5. Most participants were thus new to the staff, while the spin- and facing-driven presets are designed for trained performers.

The questionnaire was written for this pilot. It asked for agreement with six statements about the playing experience on a five-point scale (Table 4), followed by preset preference and free text. We did not use a standard DMI evaluation instrument [15, 22], and with N=10 the ratings are descriptive. Comfort scored highest, followed by novelty and absorption; predictability and expressiveness scored lower and varied more.

Preferences were spread out: *Azimut Kinetic* was chosen most often (3/10), then *Wind Noise* and *Lead+Drone* (2/10 each), and *Spin Voices (Scale)* and *Vocal Tract* once each. Reasons followed each preset's character: *Azimut Kinetic* "matches best what I know from my martial art experience", *Wind Noise* needs "no knowledge [...] in martial art or music", and *Lead+Drone* made melody easiest to steer. The tenth participant named no single preset ("All of them are great") and pointed to live patching instead ("the ability to customize live and create new patches directly inside with the node things is great"). Another asked for the wiring interface to be easier, "maybe some kind of tutorial".

The clearest finding came unprompted. Three of ten participants reported that thrusts which felt physically decisive had no matching effect on the sound ("The thrust are not well detected"; "Add 'thrust' detection support or fix the existing"). Section 7.1 gives a probable cause. Other frustration was described as learning ("when a bit hard it's just a learning experience"), and one participant found the instrument fun but not, on its own, motivating to practise.

6.4 Not Yet Measured

(i) *End-to-end latency*, to replace the estimate above: a contact microphone on the staff and the plugin output, recorded on two channels of one interface. (ii) *Packet rate, jitter, and loss*: logging inter-arrival times in the plugin. (iii) A timed, unassisted *rewiring task* (retune a parameter, then change which feature drives a sink). (iv) A replay test for the presets and a remount test for the calibration. (v) Sessions with participants trained in bô or another martial art.

7 Discussion

7.1 Limitations

Per-block analysis. The analyzer runs once per audio block on the latest packet. At 48 kHz with 128-sample buffers, a block arrives every ≈ 2.7 ms and a packet every 10 ms, so each packet is analysed about four times, and Δt comes from the host clock rather than the sensor. Every derivative-based feature (onset, jerk, tip velocity, thrust jerk) then sees one large step followed by zeros, and its magnitude depends on the buffer size. For the thrust detector, the zero steps can create a false zero-crossing on a thrust's rising edge, and its peak envelope decay (0.94, meant per packet) is applied per block. We think this explains the pilot's thrust finding. The fix is to analyse only when the packet sequence counter advances and to send the sensor's timestamp in each packet. The tuned constants and the thrust detector will then need re-tuning.

Table 3: Contributions, current evidence, and open questions.

Contribution	Current evidence	Untested
(1) bô features	implemented; spin- and facing-driven *Azimut Kinetic* was the preset most often preferred in the pilot; thrust detection failed in the pilot	facing accuracy and yaw drift against ground truth; thrust after the analyzer fix
(2) calibration	rotation independence follows from the derivation (§4.2)	remounting at several angles on hardware
(3) live graph	implemented and used throughout development; one pilot participant singled out live patching	whether other performers can rewire it unaided
(4) presets	all eleven hand-coded mappings rebuilt from one catalog (§4.5)	output equivalence with the former classes; mappings designed independently of the catalog

Table 4: Agreement on a 1–5 scale, N=10: mean (sample SD), median, and range.

Statement	Mean (SD)	Mdn	Range
Predictability ("in control, could predict")	3.4 (1.3)	4	1–5
Physical comfort	4.0 (1.1)	4	2–5
Novelty ("new ways to make sound")	3.9 (1.0)	4	2–5
Expressiveness through movement	3.3 (1.3)	3	1–5
Absorption ("less about how, more about music")	3.9 (1.0)	4	2–5
Musical satisfaction	3.6 (1.3)	4	1–5

Heading. The 6-axis Mahony filter keeps facing free of magnetic interference, but gives it no absolute reference, so yaw drift since calibration is uncorrected. We expect it to stay small over a set of a few tens of minutes but have not measured it, and there is no gesture yet to re-zero heading during a performance. The Mahony filter's fixed gain ($K_p = 30$) also treats the accelerometer as a gravity reference, which centripetal acceleration violates during fast spins; the resulting tilt error has not been measured.

Sensor position. Calibration corrects rotation but not position along the staff, so the acceleration-based features depend on where the unit is mounted (§4.3).

Network. The hotspot name, password, and destination IP are compiled into the firmware. Pairing with a different host computer, or a host whose hotspot uses a different IP, requires reflashing over USB. A laptop hotspot also limits range and is sensitive to interference. There is no discovery of the host and no feedback when joining fails.

Interface. The graph editor serves patch authors better than performers who only want to choose a preset; there is no simplified performance view.

Robustness. Incoming OSC argument types are not checked, lost or reordered packets are not detected, and the hand-off of a sensor snapshot from the network thread to the audio thread retries in a spin loop rather than reusing the last value.

Battery. Runtime on the 2000 mAh battery has not been measured.

7.2 Future Work

The immediate steps are the per-packet analyzer and the measurements listed in §6. Replacing the hotspot with ESP-NOW, sending to a second ESP32 on the host's USB port, would remove network credentials from the firmware. Longer term, we plan a dual-IMU configuration (one per end of the staff, which would also let centripetal acceleration be separated from linear motion), gesture-triggered preset switching, and a shareable patch format.

8 Conclusions

REMORA makes a bô staff into a musical instrument whose features (spin plane, spin direction, facing, lap count, thrust) come from the way a staff is handled. Its mapping is a patch that can be rewired while it plays, and the project's hand-coded mappings now live as presets of one node catalog. In a ten-person pilot, first-time players rated it comfortable and engaging and most often preferred a spin- and facing-driven preset. They also found that thrusts were poorly detected; we identify a probable cause and a fix. Next come that fix, measured latency and packet rate, a rewiring task, and sessions with trained martial artists.

Acknowledgments

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9 Supplementary Material

- **Source and design files:** firmware, plugin source, OpenSCAD enclosure, all fifteen preset graphs, and the tool that counts each preset's nodes: https://github.com/TimotheeDvt/REMORA

10 Ethical Statement

Accessibility. REMORA assumes the mobility and two-handed grip strength needed to swing a 1–1.5 kg staff, and no reduced-effort physical variant exists. The skill barrier varies across presets (*Simple* needs only tilt and roll), and a facilitator can patch a simpler mapping for a newcomer, but this does not address the physical requirement.

Environmental impact. The enclosure is attached with zip ties, so the electronics and battery can be removed and reused. No lifecycle analysis has been performed.

Socio-economic fairness. REMORA uses free and open-source tools (JUCE, PlatformIO, OpenSCAD) and inexpensive, widely available hardware, and all source and design files are public.

Human participants. The pilot (§6.3) involved ten adult volunteers who answered an anonymous questionnaire. No names or identifying data were recorded, free-text answers are quoted without attribution, and participants could stop at any time. Each participant gave verbal consent before the session.

Use of Artificial Intelligence. We disclose that a locally-run large language model (Mistral 7B) was used to help draft and edit portions of this manuscript's prose under the first author's direction and review. This disclosure applies only to the writing of this paper: REMORA's firmware, plugin, and mapping code were designed and written entirely by the first author without AI code generation.

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